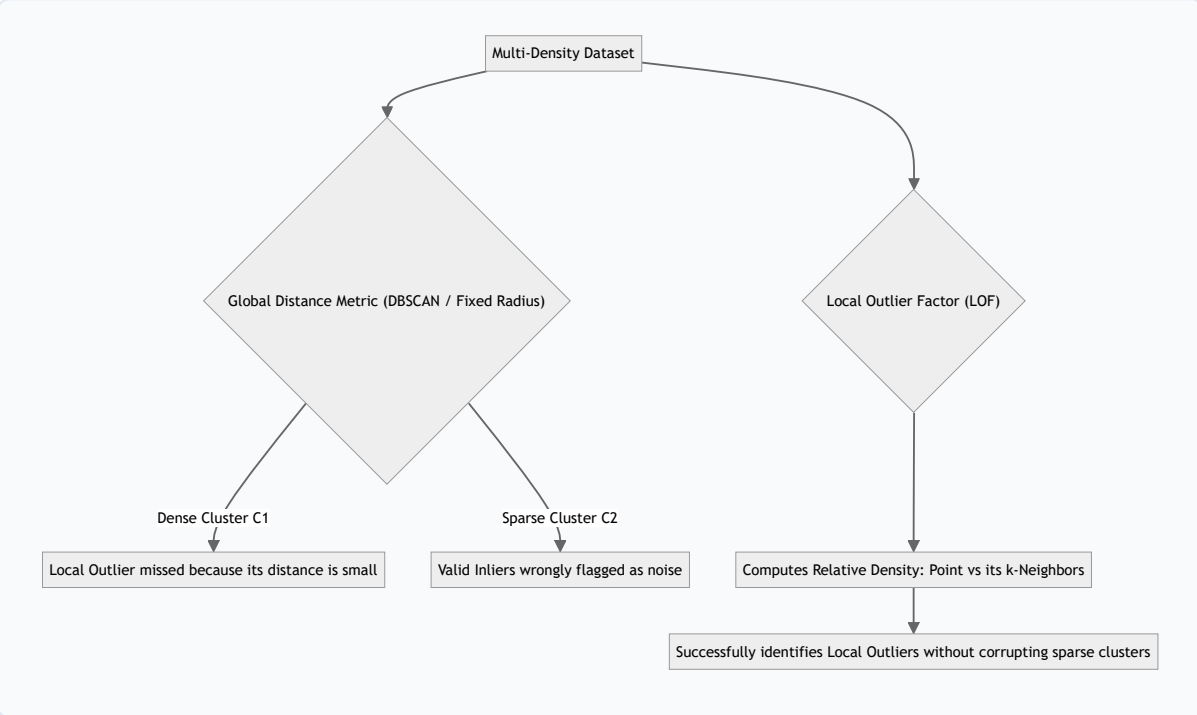
Module 32.10: Local Outlier Factor (LOF) for Anomaly Detection

1. Overview & The Problem with Global Outliers

Traditional anomaly detection algorithms (such as DBSCAN with a single global ϵ or basic distance thresholding) evaluate points against a global density standard. However, real-world datasets frequently exhibit **multi-density clusters**—regions where one cluster is extremely dense while another legitimate cluster is naturally sparse.



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2. Global vs. Local Outlier Distinction

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Feature 2
   ^
   |        (Cluster 1: Very Dense)                (Cluster 2: Sparse)
   |           * * *                                 *       *
   |          * * * *      o (Local Outlier)
   |           * * *                               *     *       *
   |
   |   o (Global Outlier)
   +-------------------------------------------------------------------> Feature 1
```

- **Global Outlier**: A data point situated far away from **all** clusters in the feature space (easily caught by DBSCAN, K-Means, or Isolation Forest).
- **Local Outlier**: A data point located in proximity to a dense cluster, but with a local density significantly lower than that of the cluster members.
    - In absolute Euclidean terms, this point's distance from Cluster 1 may be smaller than the distance between normal points inside Cluster 2!
    - A global threshold fails: if the threshold is tight, Cluster 2 is falsely destroyed as noise; if loose, the local outlier near Cluster 1 is missed.
    - **LOF solves this by evaluating density relative to the local neighborhood.**

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3. Mathematical Formulation of LOF

The LOF algorithm proceeds through four sequential mathematical computations:

Step 1: Choosing k (Neighborhood Size)

We specify hyperparameter k (the number of nearest neighbors, e.g., $k = 20$). For any point p, $N_k(p)$ represents the set of its k-nearest neighbors.

Step 2: k-Distance of Point p ($d_k(p)$)

The Euclidean distance $d(p, o)$ between p and its k-th nearest neighbor o.

Step 3: Reachability Distance

The reachability distance of point p with respect to neighbor o is defined as:

$$\text{reach-dist}_k(p, o) = \max\left(d_k(o), d(p, o)\right)$$

- If point p is sufficiently far from o, the reachability distance is simply their actual Euclidean distance $d(p, o)$.
- If p and o are very close, the distance is lower-bounded by $d_k(o)$ to stabilize statistical fluctuations in dense regions.

Step 4: Local Reachability Density (LRD)

The local reachability density of point p is the inverse of the average reachability distance over its k-nearest neighbors:

$$\text{lrd}_k(p) = \frac{|N_k(p)|}{\sum_{o \in N_k(p)} \text{reach-dist}_k(p, o)}$$

- High $\text{lrd}_k(p)$ indicates that point p is surrounded by densely packed neighbors.
- Low $\text{lrd}_k(p)$ indicates that point p resides in an isolated, sparse neighborhood.

Step 5: Local Outlier Factor (LOF Calculation)

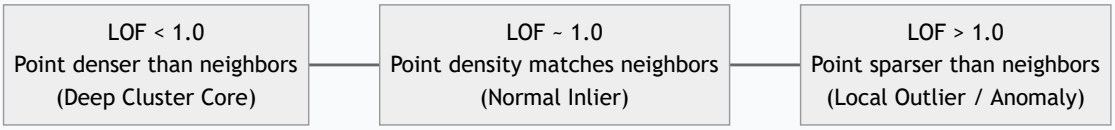
The LOF score compares the local reachability density of point p with the average local reachability density of its neighbors:

$$\text{LOF}_k(p) = \frac{\sum_{o \in N_k(p)} \frac{\text{lrd}_k(o)}{\text{lrd}_k(p)}}{|N_k(p)|} = \frac{\frac{1}{|N_k(p)|} \sum_{o \in N_k(p)} \text{lrd}_k(o)}{\text{lrd}_k(p)}$$

$$\text{LOF} = \frac{\text{Average Density of Neighbors}}{\text{Density of Point } p}$$

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4. LOF Score Interpretation & Decision Criteria



LOF Score Range	Density Comparison	Classification	Scikit-Learn Label
$\text{LOF} \approx 1$	$\text{lrd}(p) \approx \text{lrd}(\text{neighbors})$	**Normal Inlier**: Homogeneous density region	`1`
$\text{LOF} < 1$	$\text{lrd}(p) > \text{lrd}(\text{neighbors})$	**Core Inlier**: Deep inside a high-density cluster core	`1`
$\text{LOF} > 1$	$\text{lrd}(p) < \text{lrd}(\text{neighbors})$	**Local Outlier / Anomaly**: Point is isolated compared to its cluster	`-1`

LOF Score Range	Density Comparison	Classification	Scikit-Learn Label
$\text{LOF} \gg 1$	$\text{lrd}(p) \ll \text{lrd}(\text{neighbors})$	**Strong Outlier**: Extreme anomaly	`-1`

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5. Architectural Comparison: LOF vs. Isolation Forest vs. DBSCAN

Feature	DBSCAN	Isolation Forest	Local Outlier Factor (LOF)
Underlying Concept	Global density connectivity	Tree-based random isolation depth	Relative k-nearest neighbor density ratio
Local Outlier Sensitivity	Fails across multi-density clusters	Moderate (tree splits can slice across densities)	**Superior**: Specifically engineered for variable densities
Computational Complexity	$O(N^2)$ (distance matrix)	$O(t \cdot \psi \log \psi)$ (linear/sub-linear)	$O(N^2)$ (k-NN search)
Best Used When	Spatial clustering + noise removal	High-dimensional tabular datasets ($d > 20$)	Variable-density clusters in low-to-medium dimensions